ENTERPRISE NETWORK PROJECT

BGP EVPN VXLAN Fabric Implementation

Multi-tenant spine-leaf data center design, configuration and verification

Document Status	Final Engineering Documentation
Target Topology	Spine-Leaf Data Center Architecture
Platform Scope	Cisco NX-OS (Nexus 9000v) and Cisco IOS / IOS-XE (CSR1000v / IOL)
Key Features	BGP EVPN control plane • L2VNI / L3VNI • Distributed Anycast Gateway • ARP Suppression • BGP Ingress Replication

Executive Summary

This project documents the design, implementation, and verification of a multi-tenant BGP EVPN VXLAN data center fabric. The architecture moves away from traditional flood-and-learn data plane dynamics in favor of a Multi-Protocol BGP (MP-BGP) control plane, leveraging Layer-2 and Layer-3 Virtual Network Identifiers (VNIs) for multi-tenancy, Distributed Anycast Gateways for optimal default routing, and ARP Suppression to minimize broadcast traffic.

1. Physical & Logical Network Architecture

1.1 Network Topology Diagram

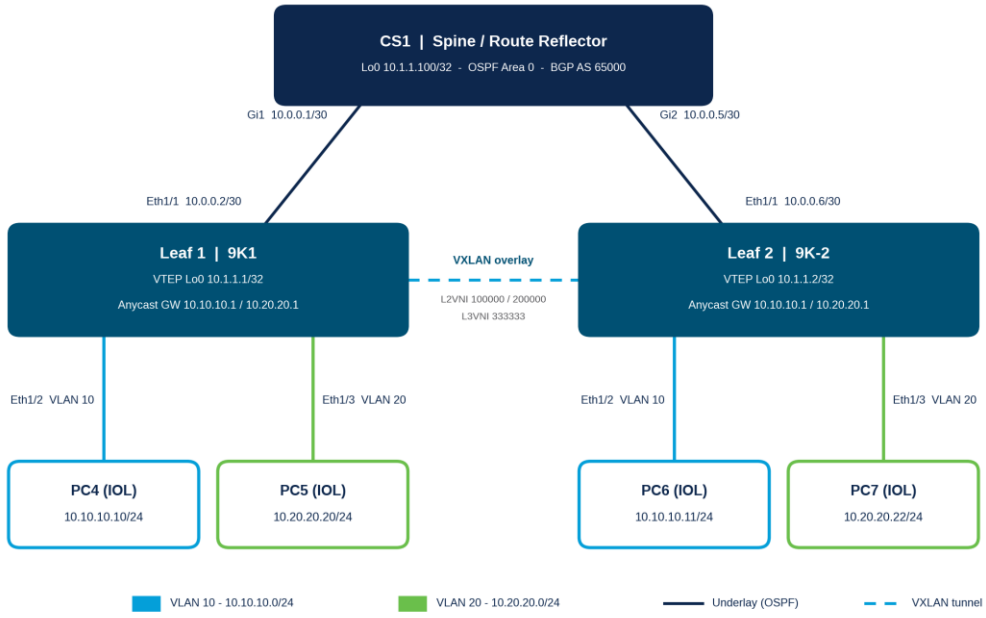
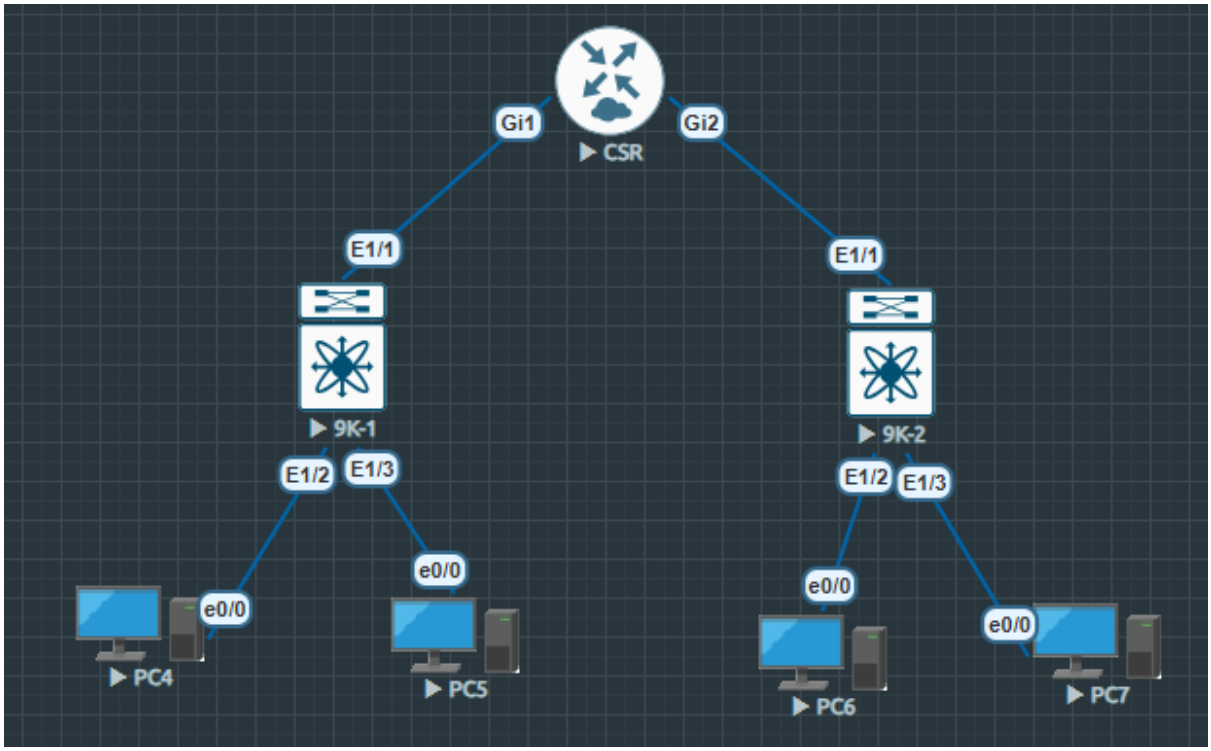


Figure 1. Spine-leaf fabric topology with underlay links, VTEPs and tenant hosts



1.2 IP Addressing & VNI Mapping

Device	Interface / Role	IP Address / Prefix	VRF	VNI	Description
CS1	Loopback0	`10.1.1.100/32`	default	—	OSPF Router ID / BGP Route Reflector ID
CS1	Gi1	`10.0.0.1/30`	default	—	Underlay link to 9K1
CS1	Gi2	`10.0.0.5/30`	default	—	Underlay link to 9K-2
9K1	Loopback0	`10.1.1.1/32`	default	—	VTEP source address / OSPF Router ID
9K1	Loopback1	`1.1.1.1/32`	default	—	Anycast / BGP identifier
9K1	Eth1/1	`10.0.0.2/30`	default	—	Point-to-point transit to CS1
9K1	Vlan10 (SVI)	`10.10.10.1/24`	CBT	100000	Distributed Anycast Gateway (Subnet A)
9K1	Vlan20 (SVI)	`10.20.20.1/24`	CBT	200000	Distributed Anycast Gateway (Subnet B)
9K1	Vlan333 (SVI)	ip forward	CBT	333333	L3 transit VNI SVI
9K-2	Loopback0	`10.1.1.2/32`	default	—	VTEP source address / OSPF Router ID
9K-2	Loopback1	`2.2.2.2/32`	default	—	Anycast / BGP identifier
9K-2	Eth1/1	`10.0.0.6/30`	default	—	Point-to-point transit to CS1
9K-2	Vlan10 (SVI)	`10.10.10.1/24`	CBT	100000	Distributed Anycast Gateway (Subnet A)
9K-2	Vlan20 (SVI)	`10.20.20.1/24`	CBT	200000	Distributed Anycast Gateway (Subnet B)
9K-2	Vlan333 (SVI)	ip forward	CBT	333333	L3 transit VNI SVI
PC4	Eth0/0 (IOL)	`10.10.10.10/24`	default	—	Host in VLAN 10 on Leaf 1
PC5	Eth0/0 (IOL)	`10.20.20.20/24`	default	—	Host in VLAN 20 on Leaf 1
PC6	Eth0/0 (IOL)	`10.10.10.11/24`	default	—	Host in VLAN 10 on Leaf 2
PC7	Eth0/0 (IOL)	`10.20.20.22/24`	default	—	Host in VLAN 20 on Leaf 2

2. VXLAN Underlay & Transport

2.1 Technical Concept

The VXLAN underlay provides high-speed, reachability-driven IP transport across the fabric. VTEPs encapsulate Layer 2 Ethernet frames into standard UDP packets (destination port 4789).

- **OSPF Area 0:** establishes point-to-point IP connectivity across the core (`10.0.0.0/30` links) and loopback addresses (`10.1.1.x/32`).
- **Head-End Ingress Replication:** BUM traffic (Broadcast, Unknown Unicast, Multicast) across same-VLAN segments is handled via BGP Ingress Replication over unicast tunnels, removing the need for PIM multicast in the underlay core.

2.2 Core Node Configurations

Spine Route Reflector (CS1) — Underlay

```
hostname CS1

interface Loopback0
 ip address 10.1.1.100 255.255.255.255
 ip ospf 1 area 0

interface GigabitEthernet1
 ip address 10.0.0.1 255.255.255.252
 ip ospf 1 area 0

interface GigabitEthernet2
 ip address 10.0.0.5 255.255.255.252
 ip ospf 1 area 0

router ospf 1
```

Leaf 1 (9K1) — NVE Tunnel Setup

```
feature nv overlay
feature vn-segment-vlan-based

interface loopback0
 ip address 10.1.1.1/32
 ip router ospf 1 area 0.0.0.0

interface nve1
 no shutdown
 host-reachability protocol bgp
 source-interface loopback0
 member vni 100000
   suppress-arp
   ingress-replication protocol bgp
 member vni 200000
   suppress-arp
   ingress-replication protocol bgp
 member vni 333333 associate-vrf
```

> **Note** Leaf 2 (`9K-2`) uses the same NVE configuration with `loopback0` set to `10.1.1.2/32`.

3. EVPN Control Plane (MP-BGP)

3.1 Technical Concept

Multi-Protocol BGP with the `l2vpn evpn` address family serves as the central control plane. Instead of learning host locations via data-plane flooding, VTEPs advertise host reachability through standardized EVPN route types:

- **EVPN Route Type 2:** advertises host MAC and IP address bindings across VTEPs.
- **EVPN Route Type 5:** advertises external or inter-VRF prefix routes for routing outside the local segment.

3.2 BGP Control Plane Configurations

Route Reflector (CS1)

```
router bgp 65000
 bgp log-neighbor-changes
 neighbor 10.0.0.2 remote-as 65000
 neighbor 10.0.0.6 remote-as 65000

 address-family l2vpn evpn
  neighbor 10.0.0.2 activate
  neighbor 10.0.0.2 send-community both
  neighbor 10.0.0.2 route-reflector-client
  neighbor 10.0.0.6 activate
  neighbor 10.0.0.6 send-community both
  neighbor 10.0.0.6 route-reflector-client
 exit-address-family
```

VTEP Leaf (9K1 / 9K-2)

```
nv overlay evpn
feature bgp

router bgp 65000
 address-family l2vpn evpn
 neighbor 10.0.0.1
   remote-as 65000
   address-family l2vpn evpn
     send-community extended

evpn
 vni 100000 l2
   rd auto
   route-target import auto
   route-target export auto
 vni 200000 l2
   rd auto
   route-target import auto
   route-target export auto
```

> **Note** The neighbor address shown is for `9K1` (CS1 `Gi1`). On `9K-2`, use `10.0.0.5` (CS1 `Gi2`).

4. Layer-3 VNI (L3VNI) & Distributed Anycast Gateways

4.1 Technical Concept

Inter-subnet routing between tenant VLANs is accomplished via **Symmetric Integrated Routing and Bridging (IRB)** utilizing a Layer-3 transit VNI (`333333`).

- **Distributed Anycast Gateway:** the default gateway IP for VLAN 10 (`10.10.10.1`) and VLAN 20 (`10.20.20.1`) is identical across both Leaf nodes, paired with a common virtual MAC (`000a.000a.000a`). This enables seamless host mobility and optimal first-hop routing.
- **ARP Suppression:** stops local ARP broadcast request storms by having the local leaf respond to host ARP requests on behalf of remote endpoints using information learned from BGP EVPN Type-2 routes.

4.2 L3VNI & VRF Configuration

Anycast Gateway MAC, VLAN-to-VNI Mapping and Tenant VRF

```
fabric forwarding anycast-gateway-mac 000a.000a.000a

vlan 10
 vn-segment 100000
vlan 20
 vn-segment 200000
vlan 333
 vn-segment 333333

vrf context CBT
 vni 333333
 rd auto
 address-family ipv4 unicast
   route-target both auto
   route-target both auto evpn
```

Anycast Gateway SVIs and L3 Transit SVI

```
interface Vlan10
 no shutdown
 vrf member CBT
 ip address 10.10.10.1/24
 fabric forwarding mode anycast-gateway

interface Vlan20
 no shutdown
 vrf member CBT
 ip address 10.20.20.1/24
 fabric forwarding mode anycast-gateway

interface Vlan333
 no shutdown
 vrf member CBT
 ip forward
```

BGP VRF Route Advertisement

```
route-map PERMIT_ALL permit 10

router bgp 65000
 vrf CBT
   address-family ipv4 unicast
     redistribute direct route-map PERMIT_ALL
```

5. Edge Host Configurations (IOL Routers)

To ensure reliable network testing and stack debugging, Cisco IOS-on-Unix (IOL) Layer 3 instances were deployed as endpoints instead of basic VPCS nodes.

```
! Host PC4 (VLAN 10 - Leaf 1)
hostname PC4
interface Ethernet0/0
 ip address 10.10.10.10 255.255.255.0
 no shutdown
ip route 0.0.0.0 0.0.0.0 10.10.10.1

! Host PC5 (VLAN 20 - Leaf 1)
hostname PC5
interface Ethernet0/0
 ip address 10.20.20.20 255.255.255.0
 no shutdown
ip route 0.0.0.0 0.0.0.0 10.20.20.1
```

```
! Host PC6 (VLAN 10 - Leaf 2)
hostname PC6
interface Ethernet0/0
 ip address 10.10.10.11 255.255.255.0
 no shutdown
ip route 0.0.0.0 0.0.0.0 10.10.10.1

! Host PC7 (VLAN 20 - Leaf 2)
hostname PC7
interface Ethernet0/0
 ip address 10.20.20.22 255.255.255.0
 no shutdown
ip route 0.0.0.0 0.0.0.0 10.20.20.1
```

6. Verification & Proof of Concept

6.1 Control Plane Verification

1. Verify EVPN Route Exchange — `show bgp l2vpn evpn vrf CBT`

```
9K1# show bgp l2vpn evpn vrf CBT
BGP routing table entry for [2]:[0]:[0]:[48]:[5000.0001.0000]:[32]:[10.10.10.11]/152, version 18
Paths: (1 available, best #1)
Flags: (0x000202) on Tree - internal ip
  VPN Routing Table -- CBT
  10.1.1.2 (metric 3) from 10.0.0.1 (10.1.1.100)
    Origin IGP, metric 0, localpref 100, valid, internal, best
    EVPN ESI: 0000.0000.0000.0000.0000, VNI: 100000, Encapsulation: VXLAN
```

Analysis Confirms Type-2 (MAC/IP) reachability for remote host `10.10.10.11` via VTEP `10.1.1.2`.

2. Verify NVE Tunnel & VNI Status — `show nve vni`

```
9K1# show nve vni
Codes: CP - Control Plane, DP - Data Plane, UP - Up, DN - Down
Interface  VNI        Multicast-group  State  Mode  Type     Cfg_mode
---------  ---------  ---------------  -----  ----  -------  --------
nve1       100000     Unicast          UP     CP    L2 [10]  CLI
nve1       200000     Unicast          UP     CP    L2 [20]  CLI
nve1       333333     Unicast          UP     CP    L3 [CBT] CLI
```

Analysis Both L2 VNIs (`100000`, `200000`) and the L3 transit VNI (`333333`) are in an `UP` state using Control Plane (`CP`) mode.

3. Verify EVPN MAC "Phonebook" & ARP Table — `show ip arp vrf CBT & show l2route evpn mac all`

```
9K1# show ip arp vrf CBT
IP ARP Table for context CBT
Total number of entries: 2
Address         Age        MAC Address    Interface      Flags
10.20.20.20     00:01:26   aabb.cc00.5000 Vlan20
10.10.10.10     00:02:10   aabb.cc00.4000 Vlan10

9K1# show l2route evpn mac all
Topology    Mac Address    Prod    Flags         Seq No     Next-Hops
----------- -------------- ------ ------------- ---------- ----------------
10          aabb.cc00.6000 BGP    Spl           0          10.1.1.2
20          aabb.cc00.7000 BGP    Spl           0          10.1.1.2
333         5000.0003.0007 VXLAN  Rmac          0          10.1.1.2
```

- **Analysis** **EVPN MAC Table (`show l2route evpn mac all`)**: Acts as the fabric's EVPN "phonebook." It explicitly proves that remote host MAC addresses (`aabb.cc00.6000` on topology 10 and `aabb.cc00.7000` on topology 20) were dynamically learned via **BGP**, resolving directly to remote VTEP `10.1.1.2` (`9K-2`) without data-plane flooding.

6.2 Data Plane End-to-End Verification

Test 1: Intra-Subnet / Same VLAN Across VTEPs (PC4 → PC6)

```
PC4# ping 10.10.10.11
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.11, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 7/7/8 ms
```

PASS 100% success via ARP Suppression and L2 VNI `100000`.

Test 2: Inter-Subnet / Cross-VLAN Across Fabric (PC5 → PC7)

```
PC5# ping 10.20.20.22
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.20.20.22, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/9/11 ms
```

PASS 100% success via Distributed Anycast Gateway and L3 VNI `333333`.

7. Troubleshooting Lessons Learned

1. **Multicast underlay removal:** when removing underlay multicast (`mcast-group 239.x.x.x`), same-VLAN ARP requests will fail unless BGP Ingress Replication (`ingress-replication protocol bgp`) is explicitly enabled on the NVE interface.
2. **ARP suppression requirement:** enabling ARP suppression (`suppress-arp`) under each member VNI allows the local leaf switch to serve host ARP requests directly from its EVPN table, eliminating broadcast storms across the fabric.
3. **Endpoint stack consistency:** using IOL router images instead of basic VPCS nodes ensures reliable ARP handling and proper TCP/IP behavior during edge verification.